



Spatial Suitability Analysis for Sustainable Camping Sites in Knuckles, Sri Lanka

DS4012 - Spatial Data Analysis

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01. Introduction

The Knuckles Conservation Forest, situated across the Kandy and Matale districts in the Central Province of Sri Lanka, is a vital ecological landscape recognized as a UNESCO World Heritage Site since 2010. Spanning approximately 175 square kilometers with elevations ranging from 900 to 1,863 meters above sea level, the range derives its name from its distinctive ridgeline profile, which resembles a clenched fist when viewed from Kandy. Defined by rugged mountain terrain, montane cloud forests, riverine ecosystems, patana grasslands, and cascading waterfalls, the region receives between 2,000 and 3,500 millimeters of annual rainfall delivered through two distinct monsoons. This unique microclimate supports exceptional biodiversity, serving as a sanctuary for species such as leopards, sambar deer, wild boar, and over 200 endemic bird species.

Growing interest in ecotourism and wilderness camping across the Knuckles range has placed increasing pressure on its fragile ecosystems. Although camping is considered a low-impact activity in protected landscapes, establishing sites without rigorous, spatial planning can cause severe environmental damage. Unmanaged campsites frequently lead to accelerated soil erosion, loss of sensitive vegetation, disruption of wildlife habitats, and pollution of freshwater streams. To prevent long-term ecological degradation while accommodating sustainable tourism, it is essential to establish an evidence-based site selection model tailored to the area's physical and biological realities.

This study integrates Geographic Information Systems (GIS) with Multi-Criteria Decision Analysis (MCDA), employing the Analytic Hierarchy Process (AHP) to evaluate the spatial suitability for campsite development in the Knuckles Conservation Forest. The methodology builds on the analytical framework established by Wang et al. (2016) for China's Kuerdening scenic area, adapted specifically to mirror Sri Lanka's environmental characteristics and conservation standards. Eight spatial variables categorized across four main criterion groups are evaluated using a three-level suitability scale, where a score of 1 indicates suitable, 2 indicates moderately suitable requiring further on-site verification, and 3 indicates unsuitable. Data retrieval and spatial processing were conducted using open-source QGIS software along with plugins like QuickOSM and SRTM Downloader, delivering a transparent and fully reproducible spatial workflow.

1.1 Spatial Nature of the Problem and Justification for GIS

Determining campsite suitability across a complex landscape is a fundamentally spatial challenge. The factors governing suitability such as terrain slope, vegetation density, proximity to streams, distance to scenic landmarks, road accessibility, mobile signal coverage, landslide risk, and flood exposure vary continuously across geographic space. Evaluating these criteria in isolation is insufficient, as suitability is defined by how these spatial conditions intersect at specific

locations. A site might feature a gentle, ideal slope yet remain unsuitable because it sits inside an active flood plain or a high-risk landslide zone. Tabular systems or standard spreadsheet models cannot analyze these overlapping spatial relationships, measure geographic buffers, or process terrain models.

Geographic Information Systems (GIS) provide the analytical framework required to integrate, process, and evaluate these multi-layered spatial datasets. For this study, GIS serves as the core platform for several practical reasons:

- **Multi-Layer Integration:** GIS enables eight distinct spatial layers, each representing an individual environmental or infrastructural criterion, to be combined into a single, comprehensive suitability model across the entire study area.
- **Spatial Precision:** Utilizing the local coordinate reference system (EPSG:5234 - Kandawala / Sri Lanka Grid) ensures meter-level accuracy when calculating distance buffers, such as stream protection zones or network coverage radii.
- **AHP Weight Application:** Mathematical criteria weights derived through the Analytic Hierarchy Process (AHP) can be applied directly to spatial data layers, calculating a definitive suitability score for every individual geographic unit across the forest reserve.
- **Terrain and Distance Processing:** GIS tools allow for direct derivation of slope profiles from Digital Elevation Models (DEMs) and enable spatial proximity calculations across rivers, roads, and key points of interest.
- **Open-Access Reproducibility:** By executing the workflow entirely within QGIS using freely available plugins (such as QuickOSM and SRTM Downloader) alongside open datasets, the analytical framework remains fully reproducible at no cost for local researchers and authorities.
- **Visual Communication:** Beyond numerical processing, GIS generates clear, publication-quality cartographic outputs that communicate suitable zones directly to forest managers, eco-tourism planners, and policy stakeholders.

1.2 Study Area

The Knuckles Conservation Forest study area, as measured from the project boundary shapefile used in this analysis, covers approximately 491.21 km² across the Kandy and Matale districts. The range is named for its unique silhouette, which resembles a clenched fist when viewed from the lowlands. This region is characterized by high annual rainfall between 2,000 mm and 3,500 mm and supports a large number of endemic plant species alongside various endangered mammals, birds, and reptiles.

The topography is rugged and steep, with elevations ranging from 900 m in the valleys to over 1,900 m at the peaks. Due to these extreme physical and ecological conditions, selecting precise locations for campsite infrastructure is essential to avoid environmental degradation.

To ensure accuracy in distance and area calculations, all spatial data in this study is processed using the **EPSG:5234 (Kandawala / Sri Lanka Grid)** coordinate reference system. This is the standard projected coordinate system for Sri Lanka, providing the precision necessary for mountain-scale mapping.

Study Area (Knuckles)

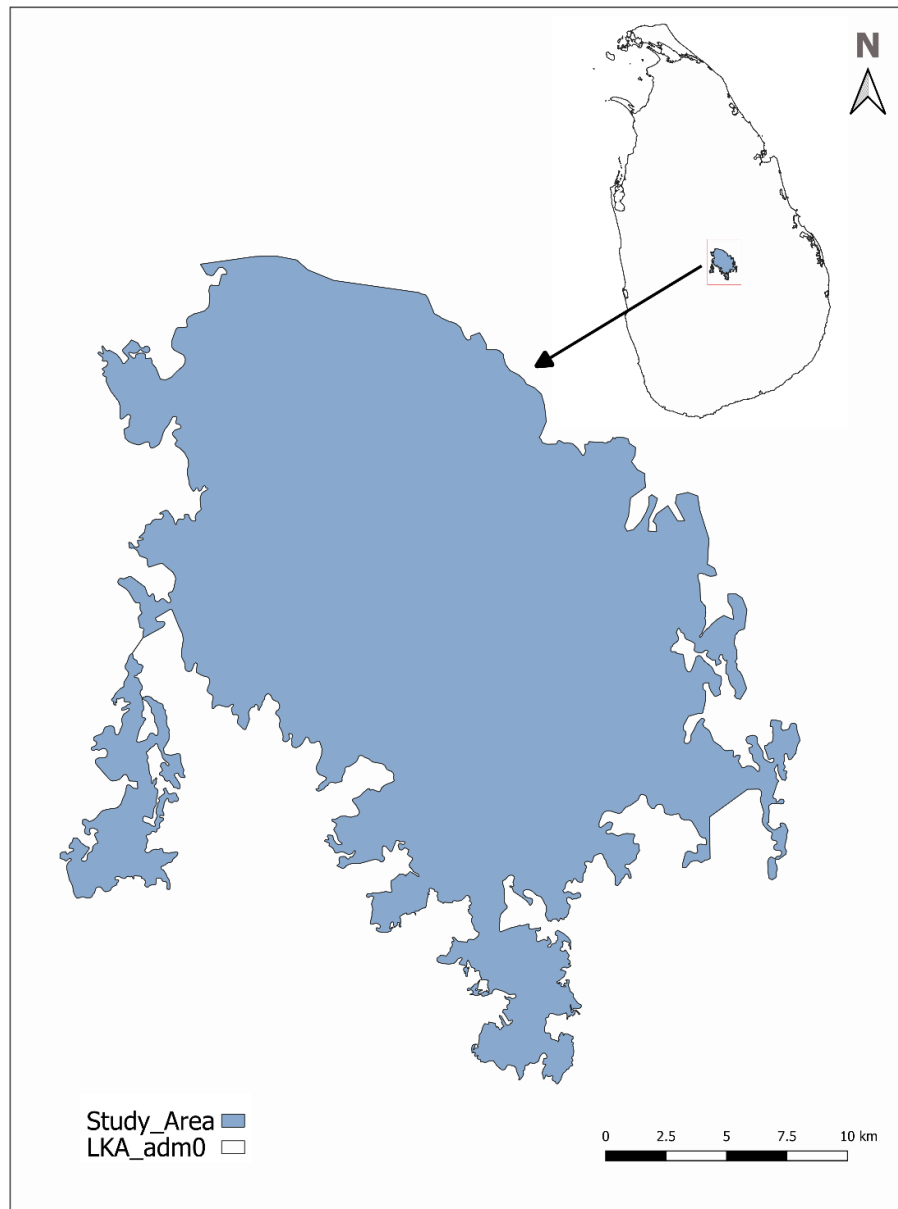


Figure 1: Study Area - Knuckles Conservation Forest

Characteristic	Details
Location	Central Province, Sri Lanka — Kandy and Matale Districts
Area	Approximately 491.21 km ² (measured from the study area boundary shapefile)
Elevation range	900 m to 1,863 m above sea level
Annual rainfall	2,000 – 3,500 mm (wet zone)
Monsoon seasons	Southwest Monsoon (May – September) and Northeast Monsoon (October – January)
UNESCO status	World Heritage Site (2010) — Central Highlands of Sri Lanka
Key habitats	Montane cloud forest, riverine forest, patana grassland, rock outcrops
Key wildlife	Leopard, wild boar, sambar deer, 200+ bird species, endemic amphibians
Primary hazards	Landslides, flash floods, forest fires on dry patana grassland
Coordinate system	EPSG:5234 — Kandawala / Sri Lanka Grid (national projected system, metres)

Table 1: Characteristics in Study Area

1.3 Objectives

The primary goal of this study is to determine the most effective locations for campsites within the Knuckles Conservation Forest through the following objectives:

1. **To identify and evaluate spatial variables** that influence campsite suitability, focusing on environmental protection and visitor safety.
2. **To process and score individual variable layers** in QGIS using spatial analysis techniques to ensure data consistency.
3. **To integrate all variable layers** through a spatial union and AHP-weighted overlay analysis to generate a final campsite suitability map.
4. **To classify the study area into three suitability levels**, identifying the highest-ranked zones as priority areas for sustainable development.

02. Methodology

This study follows a structured nine-stage GIS workflow, beginning with the definition of the decision problem and study area, proceeding through data acquisition, preprocessing, variable scoring, and Analytic Hierarchy Process weighting, and concluding with a weighted overlay, classification, and the production of a final suitability map and area statistics.

2.1 Overview of the Analytical Workflow

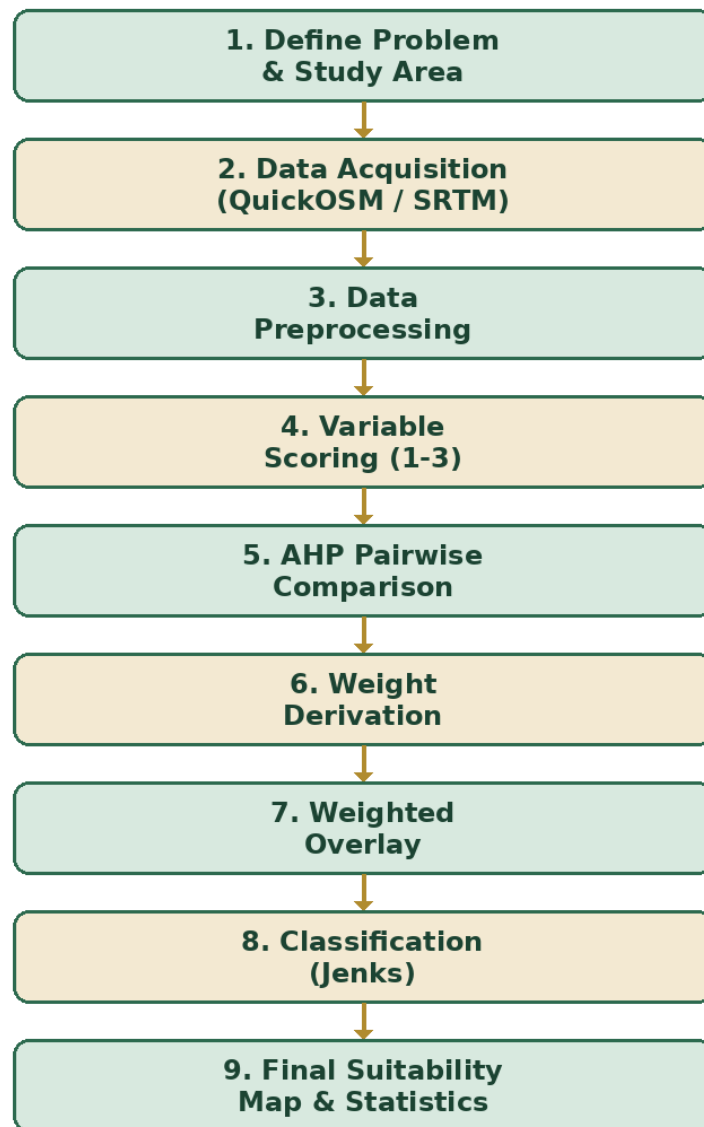


Figure 2: Methodology Workflow

2.2 Data Acquisition

All spatial data for this project is acquired and processed within QGIS using two primary open-source tools:

- **SRTM Downloader Plugin:** This tool provides 30m resolution Digital Elevation Model (DEM) data from the Shuttle Radar Topography Mission. It is used to derive topographic variables such as slope, aspect and geological disaster risks.

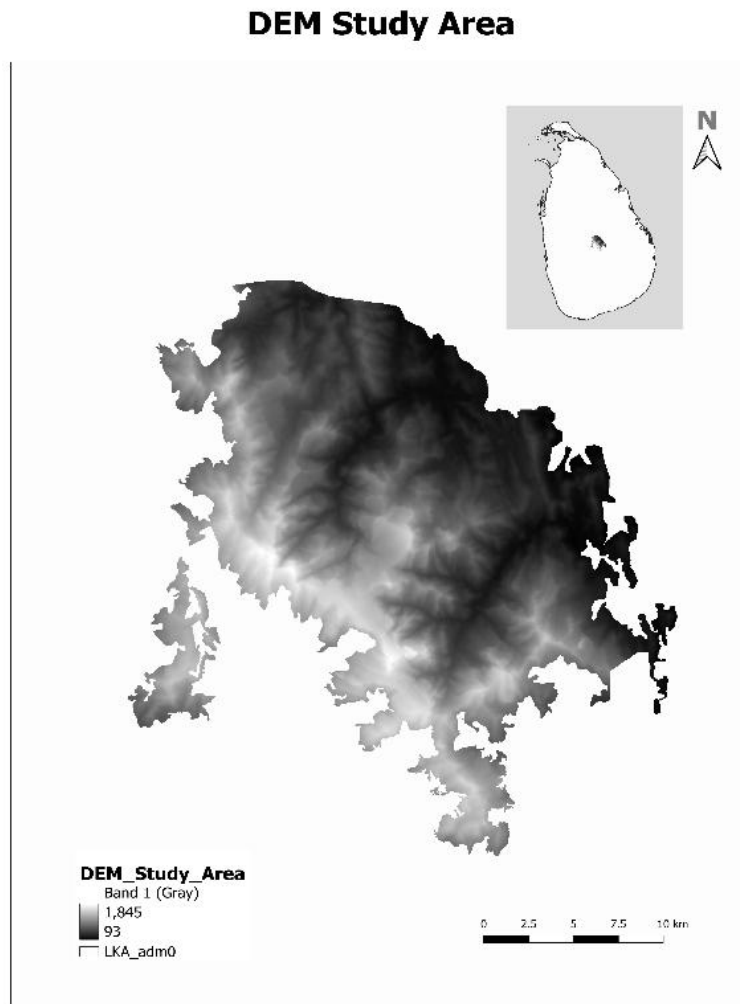


Figure 3: DEM (Digital Elevation Model) Study Area

- **QuickOSM Plugin:** This allows for the direct download of OpenStreetMap (OSM) data. It is used to extract information on land use, waterways, scenic points, tourism, roads, and infrastructure. The 'landuse' and 'natural' keys are merged during processing to ensure complete coverage of the study area.

2.3 Data Preprocessing and Cleaning

Raw spatial data acquired from open-source repositories often contains inconsistencies, mixed geometry types, or varied coordinate reference systems that must be standardized prior to analytical modeling. To ensure topological integrity, alignment, and analytical precision across all eight spatial variables, the following preprocessing steps were applied within QGIS:

- **Coordinate System Projection:** All vector data acquired via QuickOSM were reprojected from their native geographic coordinate system (EPSG:4326 - WGS84) to the local projected coordinate system (EPSG:5234 - Kandawala / Sri Lanka Grid). Similarly, the Digital Elevation Model (DEM) was reprojected to EPSG:5234 using the Warp (Reproject) tool with bilinear resampling, enforcing a uniform spatial resolution of exactly 30 meters.
- **DEM Assembly and Standardization:** Neighboring SRTM DEM tiles covering the Knuckles range were merged into a single, seamless raster grid. During reprojection to EPSG:5234, missing data pixels were explicitly assigned a NoData value of -9999 to prevent distortion during subsequent slope derivations.
- **Geometry Repair and Point Conversion:** The OpenStreetMap dataset for scenic landmarks (L4) initially contained a mixture of points, lines, and polygon features. To enable consistent distance buffering, all features were converted into single point features using the Centroids tool. Invalid spatial geometries across vector layers were subsequently passed through the Remove Null Geometries and Fix Geometries routines to eliminate self-intersecting lines and unclosed polygons.
- **Vector Layer Consolidation:** Complementary vector feature classes retrieved from separate OpenStreetMap queries were combined into unified functional layers using the Merge Vector Layers tool. This included merging land use and natural vegetation features (L2), consolidating waterways and standing water bodies (L3), and aggregating communication masts, transmission towers, and settlement centroids (L5).
- **Road Network Assembly (L6):** To map transport accessibility, comprehensive highway queries and major road networks were downloaded, clipped to the study boundary, merged, and dissolved into a continuous, non-overlapping vector line layer ready for spatial buffer generation.
- **Handling Null Values post-Union:** Following the spatial union of all eight thematic layers, polygon fragments falling outside specific buffer zones contained null entries within their respective score fields. These null values were systematically replaced with appropriate baseline scores using the `coalesce()` function in the QGIS Field Calculator before executing the final weighted suitability equations.

2.3.1 Evaluation of Technical Decisions and Preprocessing Challenges

During the spatial preprocessing and data preparation stages, several technical edge cases were encountered that directly affected the accuracy of the underlying analytical model. Resolving these challenges required deliberate parameter choices within QGIS to ensure data integrity across both raster and vector workflows.

- **Explicit Raster Cell Size Enforcement (L1 & L7):** Executing the GDAL Warp (Reproject) tool without explicit spatial limits often produces non-integer, distorted pixel dimensions (such as $30.6 \text{ m} \times 30.7 \text{ m}$) based on geographic latitude transformation. To maintain geometric precision for slope derivations and hazard modeling, the target resolution parameters were explicitly locked to $30.0 \text{ m} \times 30.0 \text{ m}$. This prevented cell misalignment and grid stretching across the reprojected Digital Elevation Model.
- **Reclassification Boundary Configuration for Flat Terrain:** The QGIS *Reclassify by Table* algorithm treats table boundary limits strictly, which can lead to edge-case errors. Standard classification tables that start at a lower bound of 0° often exclude cells with a slope of precisely 0.0° , labeling them as unclassified NoData. Because perfectly level terrain represents the primary target condition for camping grounds, the minimum boundary for the lowest suitability class was set to -1.0° . This ensured that all completely flat terrain was captured and correctly assigned a score of 1 (Suitable).
- **Raster Output Data Type and Range Bounds:** When reclassifying continuous terrain rasters into discrete suitability ranks, failing to define boundary behavior generates invalid 0-value outputs for unmapped pixels. To avoid corrupting downstream map algebra, the raster output data type was strictly constrained to 16-bit Signed Integer (Int16), and the parameter *Use NoData when no range matches value* was enabled. This prevented random 0-value noise from entering the weighted sum equations.
- **Null Value Handling in Multi-Layer Vector Union:** Consolidating eight distinct vector layers through seven sequential spatial union operations created geographic fragments where attributes from certain layers were missing, resulting in Null entries within individual score columns. Unhandled Null values automatically propagate through mathematical calculations, causing severe rendering gaps and blank polygons in the final suitability map. To solve this, default value assignments were embedded directly into the AHP scoring formula using the `coalesce()` function:
 - **Conservative Risk Default (Score 3):** For environmental and infrastructure variables, missing entries were defaulted to a score of 3 (Unsuitable) to maintain a cautious, conservation-first planning approach.
 - **Safe Boundary Default (Score 1 - L8 Flood Risk):** For the flood risk layer, Null values occurring outside the designated stream buffer zones represent high-

elevation ground safely removed from river networks, and were appropriately defaulted to a score of 1 (Suitable).

2.4 Scoring System

This project uses a three-level suitability scoring system applied consistently to all eight variables:

Score	Suitability Level	Meaning
1	Suitable	This location is suitable for campsite development
2	Moderate	This location is moderately suitable
3	Not Suitable	This location is not suitable for campsite development

Table 2: *Suitability Scoring System*

2.5 Analytical Techniques

To transform raw raster and vector spatial datasets into standardized three-level suitability scores (1 = Suitable, 2 = Moderately Suitable, 3 = Unsuitable), three distinct GIS processing techniques were applied based on the underlying structure of each variable.

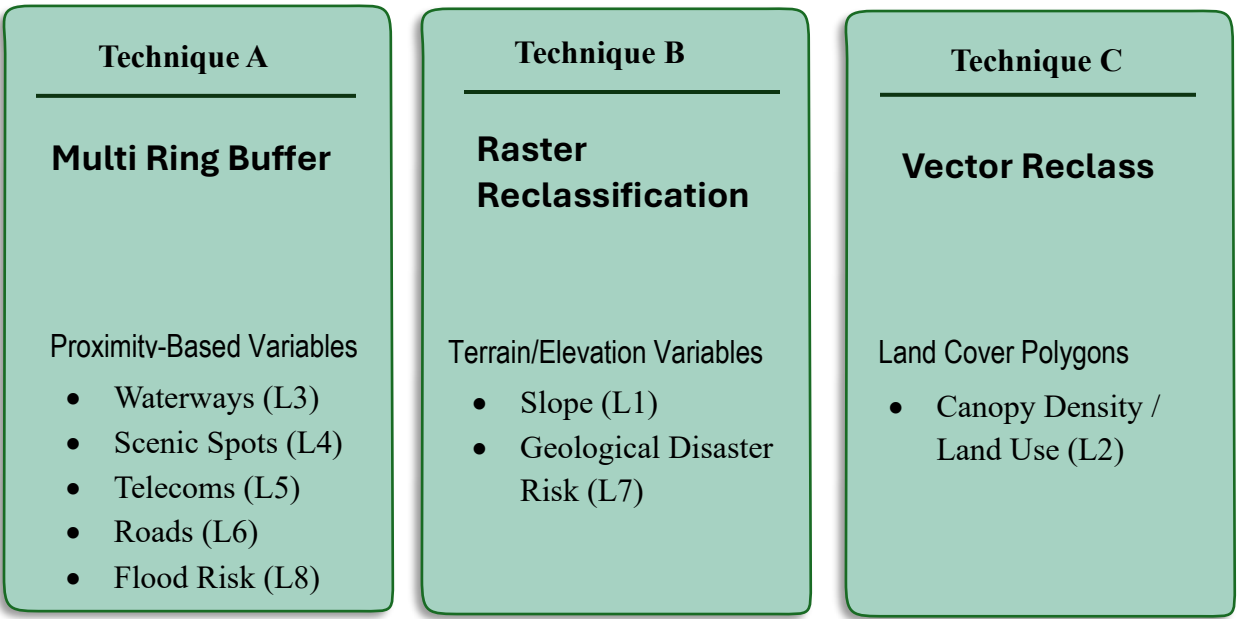


Figure 4: *Summary of the Three Analytical Techniques Applied to the Eight Variables*

2.5.1 Technique A — Multi Ring Buffer Analysis

Multi-ring buffering was applied to proximity-dependent vector variables, including distance to waterways, scenic spots, telecommunication coverage, road networks, and flood risk zones. For each feature layer, two concentric buffer distances were established, dividing the landscape into three geographic zones: an inner ring, a middle ring, and the outer area beyond the second boundary.

- **Beneficial Proximity:** For variables where close access is desirable (such as scenic landmarks, road access, and signal coverage), the inner ring was assigned a score of 1, the middle ring a score of 2, and the outer unbuffered area a score of 3.
- **Hazardous Proximity (L8 Flood Risk):** For environmental hazard exposure, this scoring logic was inverted. The high-risk inner buffer closest to river channels was assigned a score of 3 (Unsuitable), whereas the safer terrain beyond the secondary buffer received a score of 1 (Suitable).

2.5.2 Technique B — Raster Reclassification

Raster reclassification was executed on continuous terrain surfaces derived from the Digital Elevation Model, specifically slope steepness and derived geological disaster risk zones.

Using the *Reclassify by Table* tool in QGIS, continuous cell values were grouped into three distinct range intervals corresponding to the suitability scores. To maintain data integrity, outputs were strictly saved as 16-bit Signed Integers (Int16), and unmatched values were assigned as NoData to prevent flat terrain from receiving false zero-value classifications.

2.5.3 Technique C — Vector Attribute Reclassification

Vector attribute reclassification was utilized for discrete land cover and canopy density features extracted from OpenStreetMap datasets.

Because land cover types exist as categorical polygons rather than continuous surfaces or distance buffers, conditional statements (CASE WHEN) were executed within the QGIS Field Calculator. Each land use tag (such as dense forest, scrubland, or grassland) was assigned a score reflecting its underlying ecological sensitivity and physical site suitability for camping.

2.6 Variables Overview

This project evaluates eight spatial variables. Each variable is processed as a separate spatial layer and assigned scores 1, 2, or 3. The layers are combined using AHP-derived weights to produce the final suitability map.

Layer	Factor	Variable	Technique	Data Source
L1	Natural Environment	Slope	Raster Reclassify	SRTM DEM
L2	Natural Environment	Canopy Density	Vector Reclassify	QuickOSM - landuse - natural
L3	Natural Environment	Distance to Waterways	Multi Ring Buffer	QuickOSM - waterway = waterfall, rivers - natural = water
L4	Landscape Condition	Distance to Scenic Spots	Multi Ring Buffer	QuickOSM - tourism - natural = peak - waterway = waterfall
L5	Infrastructure	Telecommunication	Multi Ring Buffer	QuickOSM - man_made = mast, tower - place = town, village
L6	Infrastructure	Roads (Transportation)	Multi Ring Buffer	QuickOSM highway = primary, secondary, motorway, trunk, tertiary
L7	Safety Condition	Geological Disaster Risk	Raster Reclassify	SRTM DEM slope raster from L1
L8	Safety Condition	Flood Risk	Multi Ring Buffer	QuickOSM waterway from L3

Table 3: *Variables Overview*

2.6.1. Variable Definitions and Suitability Scoring

L1 — Slope

- ❖ **Criterion Group:** Natural Environment
- ❖ **Primary Measure:** Surface gradient and terrain steepness. Gentle, level ground is essential for pitching tents, maintaining site drainage, and allowing safe foot traffic. Highly elevated or steep slopes are physically unusable and pose severe safety hazards in mountainous terrain.
- ❖ **Justification:** Given the rugged topographies characterizing the Knuckles range, slope gradient acts as the most fundamental physical filter determining whether a location can physically support a campsite.
- ❖ **Data Source:** NASA SRTM Digital Elevation Model (30 m resolution), processed via Raster → Analysis → Slope within QGIS. This derived slope grid is also reused to compute variable L7 (Geological Disaster Risk).
- ❖ **Processing Technique:** Technique B — Raster Reclassification (grouped into three primary slope intervals).

Score	Suitability	Criteria / Range
1	Suitable	-1° to 10°
2	Moderate	10° to 20°
3	Not Suitable	20° to 90°

Table 4: L1 - Slope Suitability Criteria/Range

Technical Implementation Notes

- **Boundary Catchment Adjustment:** The lower bound of the primary class must be configured to -1.0° inside the QGIS *Reclassify by Table* interface rather than 0.0° . This avoids software boundary rounding errors and guarantees that completely flat terrain (0.0°) is correctly classified under Score 1 instead of dropped as NoData.
- **Raster Data Integrity:** The target output pixel type should be strictly set to Signed 16-bit Integer (Int16). Furthermore, enabling the *Use NoData when no range matches value* flag in the tool parameters prevents spurious 0 value pixels from populating non-target regions and skewing downstream spatial map algebra.

L1 - Slope (Raster format)

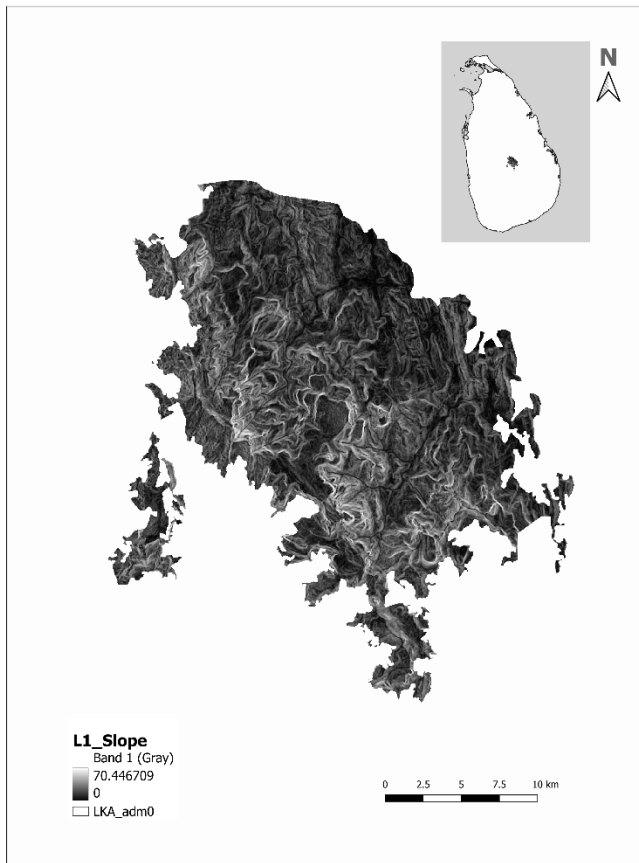


Figure 6: L1 - Slope in Raster Format

L1 - Slope (Vector format)

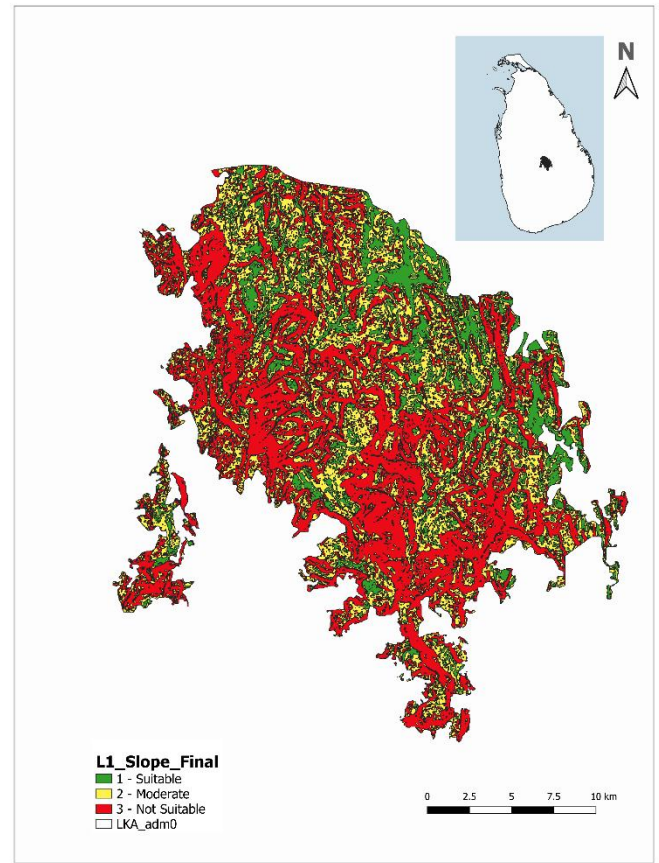


Figure 5: L1 - Slope After Raster Reclassification

fid	Slope_Score
1	3
2	2
3	2
4	3
5	3

Table 5: L1 - Slope Attribute Table (1st 5 rows)

L2 — Canopy Density

- ❖ **Criterion Group:** Natural Environment
- ❖ **Primary Measure:** Tree canopy density and vegetation structure mapped from OpenStreetMap (OSM) land use and natural cover features. Directing recreational activity toward open ground reduces habitat destruction, tree clearance, and ecological degradation in dense forest zones.

- ❖ **Justification:** The Knuckles landscape features a delicate mosaic of montane cloud forests, pyrogenic *patana* grasslands, and scrub ecosystems. Directing campsites toward open or semi-open areas minimizes physical impacts on sensitive wildlife habitats and high-value forest reserves.
- ❖ **Data Source:** QuickOSM vector datasets retrieved using key-value queries for landuse and natural, combined into a single unified polygon layer.
- ❖ **Processing Technique:** Technique C — Vector Attribute Reclassification based on land cover feature tags.

Score	Suitability	Criteria / Range
1	Suitable	natural: grassland, bare_rock, water landuse: grass, farmland
2	Moderate	natural: scrub landuse: orchard, residential
3	Not Suitable	landuse: forest natural: wood

Table 6: L2 - Canopy Density Suitability Criteria/Range

Field Calculator Expression & Technical Implementation

Vector attribute scoring was executed using the QGIS Field Calculator on the merged land use/natural polygon layer.

```

CASE
  WHEN "natural" IN ('grassland', 'bare_rock', 'water')
    OR "landuse" IN ('grass', 'farmland')
  THEN 1

  WHEN "natural" IN ('scrub')
    OR "landuse" IN ('orchard', 'residential')
  THEN 2

  WHEN "landuse" IN ('forest')
    OR "natural" IN ('wood')
  THEN 3

  ELSE 3
END

```

Figure 7: L2 - Canopy Density Field Calculator Expression

- **Conservative Conservation Default:** The ELSE 3 fallback condition ensures that any unmapped, untagged, or ambiguous polygon within a protected conservation forest is

automatically assigned a score of 3 (Unsuitable). This enforces a strict conservation-first approach, preventing unknown habitats from being incorrectly classified as safe for development.

L2 - Canopy Density

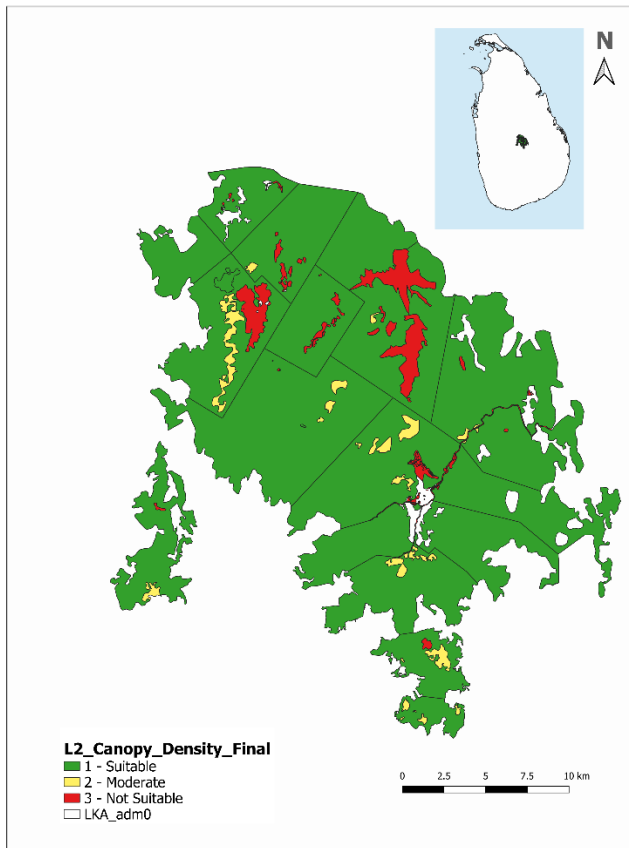


Figure 8: L2 - Canopy Density after Vector Reclassification

fid	landuse	natural	name	Canopy_Score
1	forest	NULL	NULL	1
2	forest	NULL	Knuckles Cons...	1
3	forest	NULL	NULL	1
4	forest	NULL	NULL	1
5	forest	NULL	NULL	1

Table 7: L2 - Canopy Density Attribute Table (1st 5 rows)

L3 — Distance to Waterways

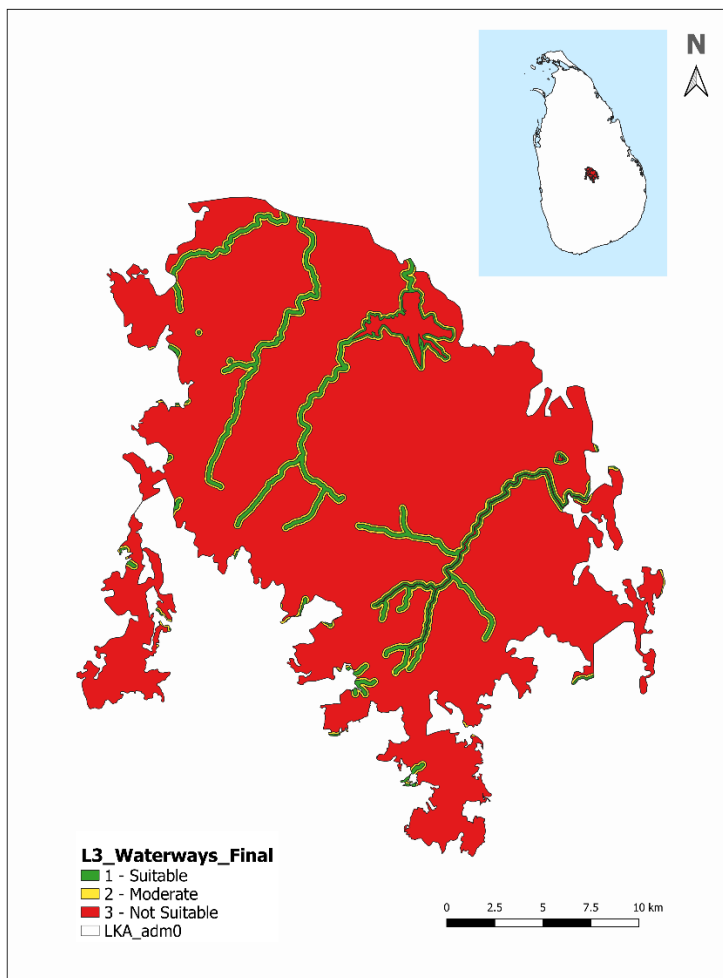
- ❖ **Criterion Group:** Natural Environment
- ❖ **Primary Measure:** Proximity to fresh surface water features, including rivers, mountain streams, waterfalls, reservoirs, and natural ponds.
- ❖ **Justification:** Reliable freshwater access is an essential logistical requirement for backcountry camping. It supports core visitor needs such as drinking water filtration, cooking, and sanitation. Locations within 100 meters of a stream provide optimal operational convenience in rugged terrain.
- ❖ **Data Source:** QuickOSM queries retrieving line features (waterway=* capturing rivers, streams, canals, and cascades) and polygon features (natural=water capturing standing water bodies), merged into a single consolidated vector dataset (L3_Waterway_OSM.shp).

- ❖ **Processing Technique:** Technique A — Multi Ring Buffer Analysis (Standard beneficial proximity modeling).

Score	Suitability	Criteria / Range
1	Suitable	0 – 100 m
2	Moderate	100 – 200 m
3	Not Suitable	> 200 m

Table 8: L3 - Waterway Suitability Criteria/Range

L3 - Waterways



fid	mrb_dist	Waterways_Score
1	200	2
2	100	1
3	NULL	3

Table 9: L3 - Waterways Attribute Table

Figure 9: L3 – Waterways in MRB

L4 — Distance to Scenic Spots

- ❖ **Criterion Group:** Landscape Condition
- ❖ **Primary Measure:** Spatial proximity to key eco-tourism points of interest, including panoramic viewpoints, prominent mountain peaks, and natural waterfalls.
- ❖ **Justification:** The primary motivation for wilderness camping in the Knuckles Conservation Forest is experiencing its world-class mountain scenery and natural landmarks. Locating campsites within reasonable hiking distance of these attractions maximizes recreational appeal and enhances overall visitor satisfaction.
- ❖ **Data Source:** QuickOSM queries retrieving tourism feature classes (tourism=viewpoint, tourism=attraction) and natural landmarks (natural=waterfall, natural=peak). All vector features are converted into uniform point geometries using the QGIS **Centroids** tool before buffering.
- ❖ **Processing Technique:** Technique A — Multi Ring Buffer Analysis (Standard beneficial proximity modeling).

Score	Suitability	Criteria / Range
1	Suitable	< 1.5 km
2	Moderate	1.5 – 3.0 km
3	Not Suitable	> 3.0 km

Table 10: L4 - Scenic Points Suitability Criteria/Range

Spatial Processing & Geometry Standardization Notes

- **Geometry Normalization via Centroids:** QuickOSM returns scenic features as a mix of point nodes, linear river cascades, and polygon boundaries. Running the **Centroids** algorithm converts all features into clean, uniform point nodes, preventing topological artifacts during multi-ring buffer calculations.
- **Buffer Rings Setup:** Two concentric buffer rings are generated at 1,500 m (1.5 km) and 3,000 m (3.0 km). Ground areas falling outside the 3,000 m threshold are assigned a score of 3 (Unsuitable) in the final attribute output.

L4 - Scenic Points

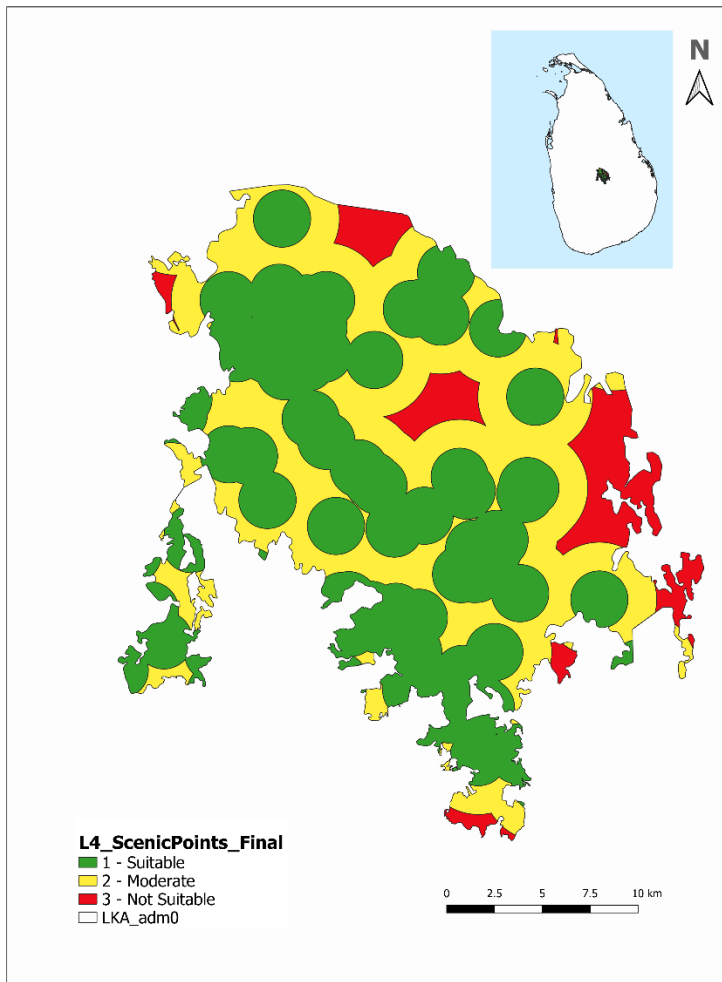


Figure 10: L4 - Scenic Points in MRB

fid	mrbs_dist	ScenicPoints_Score
1	1500	1
2	3000	2
3	1500	1
4	3000	2
5	NULL	3

Table 11: L4 - Scenic Points Attribute Table (1st 5 rows)

L5 — Telecommunication

- ❖ **Criterion Group:** Infrastructure Condition
- ❖ **Primary Measure:** Mobile communications coverage derived from proximity to cellular masts, telecom towers, and populated settlement nodes.
- ❖ **Justification:** Maintaining emergency communication channels is a baseline safety requirement for managed ecotourism sites. The rugged topography of the Knuckles range creates severe signal dead-zones; establishing campsites near known transmission infrastructure or settlement relay points ensures visitors and park rangers can initiate emergency calls when needed.
- ❖ **Data Source:** QuickOSM queries retrieving telecom infrastructure features (man_made=mast, man_made=tower) alongside settlement nodes (place=town,

place=village) that host local cellular relay antennas. All query layers are consolidated into a single point feature layer prior to spatial processing.

- ❖ **Processing Technique:** Technique A — Multi Ring Buffer Analysis (Standard beneficial proximity modeling).

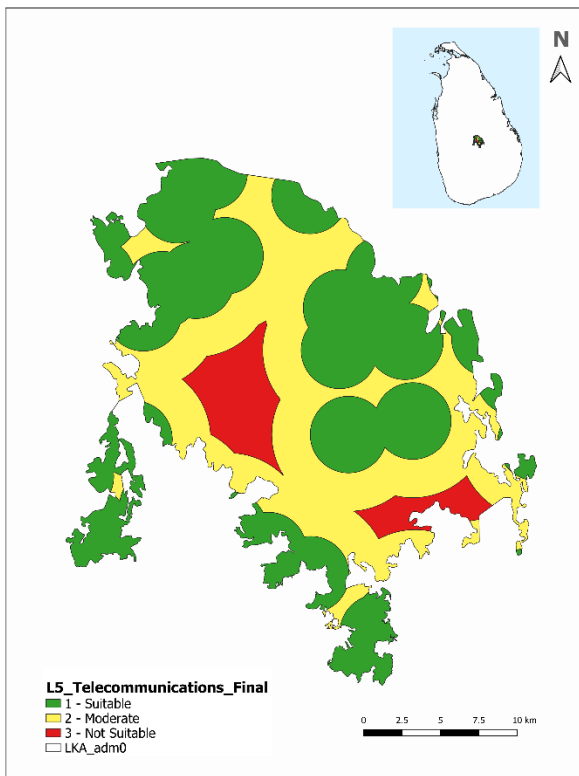
Score	Suitability	Criteria / Range
1	Suitable	0 – 2.5 km
2	Moderate	2.5 – 5.0 km
3	Not Suitable	> 5.0 km

Table 12: L5 - Telecommunication Suitability Criteria/Range

Spatial Processing & Infrastructure Modeling

- **Multi-Source Network Layering:** Combining dedicated cell towers (man_made) with settlement centroids (place) acts as a realistic proxy for network footprint, compensating for sparse open-source telemetry data in rural Sri Lanka.
- **Ring Buffer Distances:** Two concentric buffer zones are established at 2.5 km (2,500 m) and 5.0 km (5,000 m). Regions beyond 5.0 km are classified as unserved communications blackspots (*Score* = 3).

L5 - Telecommunications



fid	mr_b_dist	Telecommunications_Score
1	2500	1
2	5000	2
3	2500	1
4	5000	2
5	NULL	3

Table 13: L5 - Telecommunication Attribute Table (1st 5 rows)

Figure 11: L5 - Telecommunication in MRB

L6 — Roads (Transportation)

- ❖ **Criterion Group:** Infrastructure Condition
- ❖ **Primary Measure:** Spatial proximity to transportation networks, including all classes of motorable roads, tracks, and primary footpaths.
- ❖ **Justification:** Reliable transport access is fundamental for establishing and operating a official campsite. Proximity to roads ensures smooth visitor arrival, enables initial site construction and routine supply deliveries, and provides critical access routes for search and rescue operations or emergency medical evacuations.
- ❖ **Data Source:** QuickOSM vector datasets retrieved via a comprehensive query (key=highway with an empty value to download motorways, primary, secondary, tertiary roads, tracks, and footpaths) alongside a targeted major roads query. Both queries were clipped to the study boundary, merged, and dissolved into a single continuous network layer.
- ❖ **Processing Technique:** Technique A — Multi Ring Buffer Analysis (Standard beneficial proximity modeling)

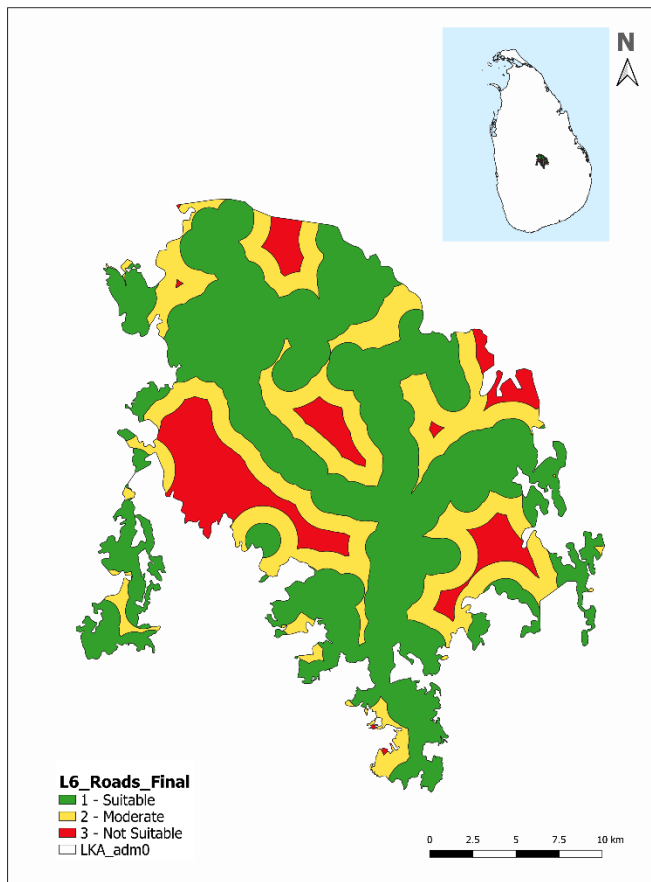
Score	Suitability	Criteria / Range
1	Suitable	0 – 1 km
2	Moderate	1 – 2 km
3	Not Suitable	> 2 km

Table 14: L6 - Roads Suitability Criteria/Range

Spatial Processing & Highway Network Preparation

- **Network Dissolve Operations:** Merging distinct OpenStreetMap highway queries often produces overlapping line segments and split nodes. Passing the merged dataset through the **Dissolve** tool creates a single multi-part geometry line feature, eliminating interior boundary intersections prior to buffering.
- **Ring Buffer Distances:** Concentric ring boundaries are established at 1,000 m(1.0 km) and 2,000 m(2.0 km). Remote forest areas situated beyond 2.0 km from the nearest transportation line are assigned a score of 3 (Unsuitable).

L6 - Roads



fid	mrbs_dist	Roads_Score
1	1000	1
2	2000	2
3	NULL	3

Table 15: L6 – Roads Attribute Table

Figure 12: L6 - Roads in MRB

L7 — Geological Disaster Risk

- ❖ **Criterion Group:** Safety Condition
- ❖ **Primary Measure:** Susceptibility to slope failure, rockfalls, and landslides based on surface gradient and topographic steepness.
- ❖ **Justification:** The Knuckles Conservation Forest is characterized by deeply dissected terrain, high relief, and intense monsoon rainfall exceeding 3,500 mm annually. Slope instability and mass movement represent the single most severe physical hazard to human safety in this landscape. Avoiding unstable, highly inclined slopes is critical for protecting campsite visitors and infrastructure.
- ❖ **Data Source:** NASA SRTM Digital Elevation Model (30 m resolution). This variable reuses the exact slope raster derived for Variable L1 (Raster → Analysis → Slope), requiring no additional data acquisition.
- ❖ **Processing Technique:** Technique B — Raster Reclassification (applying hazard-specific slope threshold intervals).

Score	Suitability	Criteria / Range
1	Suitable	Slope -1° to 15°
2	Moderate	Slope 15° to 30°
3	Not Suitable	Slope > 30°

Table 16: *L7 - Geo Disaster Suitability Criteria/Range*

Dual-Variable Comparison & Technical Setup

- **Functional Distinction (L1 vs. L7):** Although both variables share the same underlying Digital Elevation Model and slope raster, they assess distinct real-world conditions:
 - **Variable L1 (Comfort & Usability):** Focuses on micro-topography suited for pitching tents and walking (0--10° = Optimal).
 - **Variable L7 (Geological Safety):** Focuses on macro-scale structural stability and mass wasting limits (0--15° = Low Risk).
- **Reclassification Settings:** As with L1, the lowest class boundary must be set to -1.0° in the QGIS *Reclassify by Table* tool to capture 0.0° flat ground. The output pixel type must be designated as Signed 16-bit Integer (Int16) with unmatched values treated as NoData to prevent erroneous zero-value classes from distorting downstream model calculations.

L7 - Geo Disaster (Raster format)

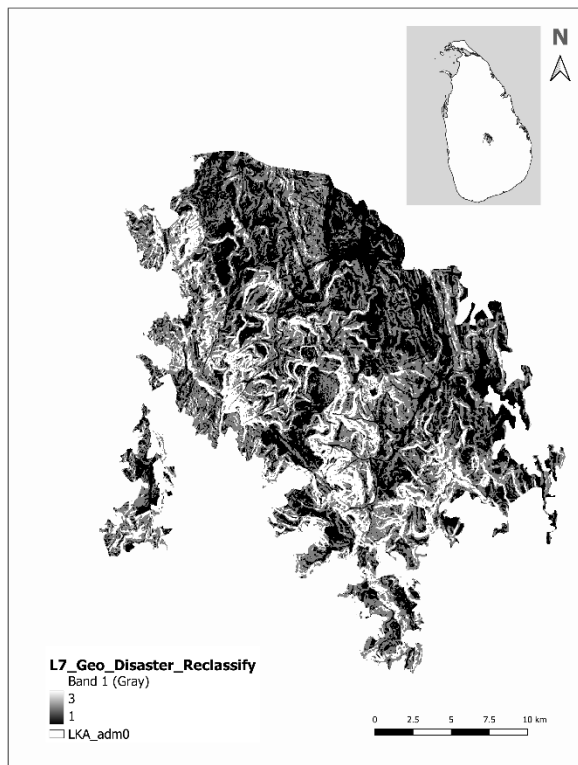


Figure 13: *L7 - Geo Disaster in Raster Format*

L7 - Geo Disaster (Vector format)

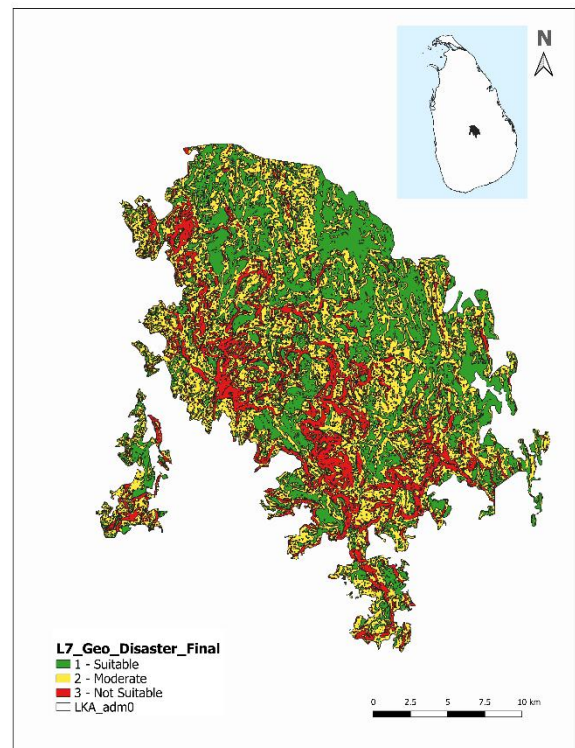


Figure 14: *L7 - Geo Disaster After Raster Reclassification*

fid	GeoDisaster_Score
1	2
2	2
3	2
4	2
5	2

Table 17: **L7 - Geo Disaster Attribute Table**
(1st 5 rows)

L8 — Flood Risk

- ❖ **Criterion Group:** Safety Condition
- ❖ **Primary Measure:** Exposure to flash flooding and stream inundation during heavy monsoonal precipitation.
- ❖ **Justification:** With annual rainfall ranging between 2,000 and 3,500 mm across two monsoon seasons, mountain streams in the Knuckles catchment experience rapid, unpredictable flash floods. Locating campsites directly along active stream banks or floodplains poses severe risks to life and property. Establishing a mandatory safety buffer away from water channels is vital for visitor safety.
- ❖ **Data Source:** Reuses the vector shapefile (L3_Waterway_OSM.shp) compiled for Variable L3, containing both linear drainage channels and standing water bodies.
- ❖ **Processing Technique:** Technique A — Multi Ring Buffer Analysis with **Reversed Scoring Logic** (where greater distance increases suitability).

Score	Suitability	Criteria / Range
3	Not Suitable	0 – 100 m
2	Moderate	100 – 200 m
1	Suitable	> 200 m

Table 18: **L8 - Flood Risk Suitability Criteria/Range**

Spatial Processing & Inverted CASE Expression

To execute the reversed logic on the multi-ring buffer output in QGIS, the following expression was evaluated in the Field Calculator:

```

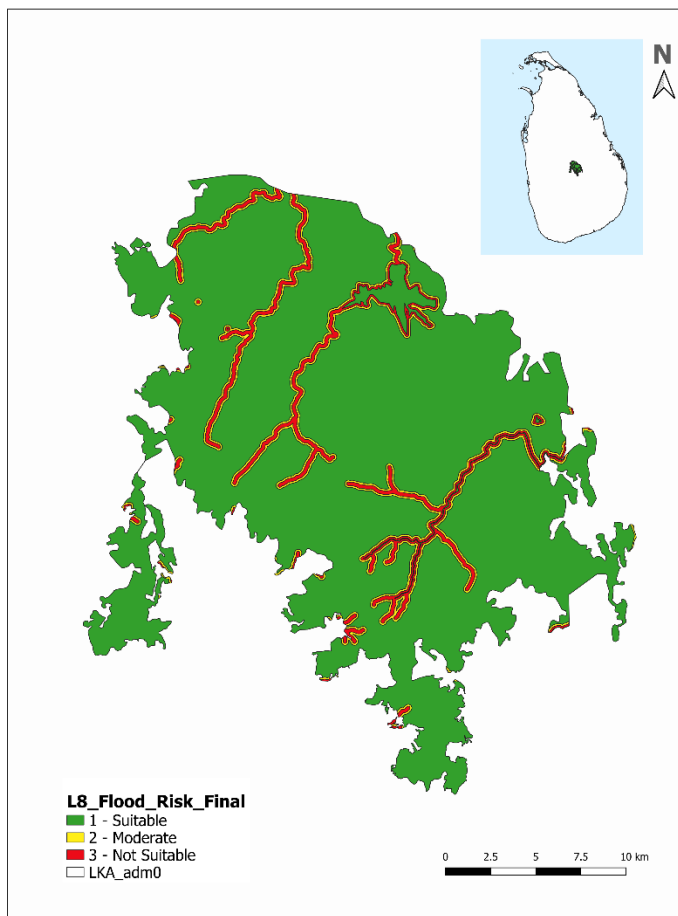
CASE
  WHEN "ringid" = 1 THEN 3
  WHEN "ringid" = 2 THEN 2
  ELSE 1
END

```

Figure 15: L8 - Flood Risk Field Calculator Expression

- **Inverted Proximity Relationship:** Unlike Variable L3 (Waterway Access), which rewards close proximity with a Score 1, L8 assigns a Score 3 to the innermost ring (0--100 m) because of severe inundation hazards.
- **Default Area Safety:** Regions situated beyond 200 m (ELSE 1) fall outside the active river overflow buffer and are categorized as safe (*Score* = 1)

L8 - Flood Risk



fid	mrbs_dist	Flood_Score
1	200	2
2	100	3
3	NULL	1

Table 19: L8 - Flood Risk Attribute Table

Figure 16: L8 - Flood Risk in MRB

03. AHP Pairwise Comparison

3.1 Analytic Hierarchy Process (AHP) Weight Derivation

The Analytic Hierarchy Process (AHP), developed by Thomas Saaty, was used to determine the relative importance of each spatial variable. Rather than assigning weights arbitrarily, AHP uses structured pairwise comparisons, evaluating two factors at a time using a standardized 1 to 9 scale.

3.1.1 Saaty's Pairwise Comparison Scale

Every criterion was compared against every other criterion based on the physical and environmental priorities of the Knuckles Conservation Forest.

Scale value	Meaning	Example from this study
1	Equally important	Telecommunication and Canopy Density are both minor factors.
3	Moderately more important	Scenic Spots is moderately more important than Slope.
5	Strongly more important	Geological Disaster Risk is strongly more important than Scenic Spots.
7	Very strongly more important	Geological Disaster Risk is very strongly more important than Roads.
9	Extremely more important	Safety-related factors compared with minor infrastructure factors.
2, 4, 6, 8	Intermediate values	Used for judgements that fall between the main scale values.

Table 20: Saaty's Pairwise Comparison Scale

3.1.2 Matrix Construction and Mathematical Normalization

To calculate the weights for $n = 8$ variables, a 8×8 comparison matrix A was constructed. The diagonal elements are set to 1.00 (comparing a variable to itself). The upper right triangle contains the relative importance judgments, and the lower left triangle contains their reciprocal values ($a_{ji} = 1/a_{ij}$).

Step 1: Complete Pairwise Comparison Matrix

Variable	L1	L2	L3	L4	L5	L6	L7	L8
L1	1.000	2.269	1.873	0.715	2.145	2.458	0.387	0.608
L2	0.441	1.000	0.825	0.315	0.945	1.083	0.170	0.268
L3	0.534	1.212	1.000	0.382	1.145	1.313	0.207	0.325
L4	1.399	3.173	2.619	1.000	3.000	3.438	0.541	0.851
L5	0.466	1.058	0.873	0.333	1.000	1.146	0.180	0.284
L6	0.407	0.923	0.762	0.291	0.873	1.000	0.157	0.247
L7	2.584	5.865	4.841	1.848	5.545	6.354	1.000	1.572
L8	1.645	3.731	3.079	1.175	3.527	4.042	0.636	1.000
Column Total (Σ)	8.476	19.231	15.872	6.059	18.180	20.834	3.278	5.155

Table 21: Complete Pairwise Comparison Matrix Table between All Variables

L1 – Slope

L2 – Canopy Density

L3 – Waterways

L4 – Scenic Points

L5 – Telecommunication

L6 – Roads

L7 – Geo Disaster

L8 - Flood Risk

Step 2: Column Normalization & Average Row Calculation (Weight W_i)

Each cell in the matrix is divided by its respective column total:

$$\text{Normalized Cell } \bar{a}_{ij} = \frac{a_{ij}}{\sum_{k=1}^n a_{kj}}$$

The final weight W_i for each criterion is calculated by taking the average across the normalized row:

$$W_i = \frac{1}{n} \sum_{j=1}^n \bar{a}_{ij}$$

- **Calculation Example for L7 (Geological Hazard):**
 - Column 1 Contribution: $2.584/8.476 = 0.3048$
 - Column 2 Contribution: $5.865/19.231 = 0.3050$
 - Column 7 Contribution: $1.000/3.278 = 0.3051$
 - Row Average (W_7): $(0.3048 + 0.3050 + \dots + 0.3051)/8 = \mathbf{0.305}$ (or **30.5%** of the overall decision weight).

3.1.3 Final Derived AHP Weights Summary Table

Criterion Group	Layer Code	Spatial Variable	Calculated AHP Weight	Weight %	Priority Hierarchy
Safety Condition	L7	Geological Disaster Risk	0.305	30.5%	Rank 1 (Highest)
Safety Condition	L8	Flood Risk	0.194	19.4%	Rank 2
Landscape Condition	L4	Distance to Scenic Spots	0.165	16.5%	Rank 3
Natural Environment	L1	Slope	0.118	11.8%	Rank 4
Natural Environment	L3	Distance to Waterways	0.063	6.3%	Rank 5
Infrastructure Condition	L5	Telecommunication	0.055	5.5%	Rank 6
Natural Environment	L2	Canopy Density	0.052	5.2%	Rank 7
Infrastructure Condition	L6	Roads (Transportation)	0.048	4.8%	Rank 8 (Lowest)
Total Model Weight			1.000	100.0%	

Table 22: *AHP (Analytic Hierarchy Process) Weights Summary Table*

Key Insight: Safety factors (**L7 + L8 = 0.499**) account for **49.9%** (nearly half) of the entire model weight. This mathematically ensures that in a high-rainfall, steep mountain landscape like the Knuckles, life safety strictly dominates scenic or logistical convenience.

3.1.4 Consistency Ratio (CR) Validation

To ensure that the pairwise judgments are logically sound and free from contradictions, a mathematical consistency check was performed.

1. **Calculate Vector ($A \cdot W$):** Multiply the pairwise comparison matrix A by the weight vector W .
2. **Calculate Principal Eigenvalue ($\lambda_{\max}$):**

$$\lambda_{\max} = \frac{1}{n} \sum_{i=1}^n \frac{(A \cdot W)_i}{W_i} = 8.084$$

3. **Calculate Consistency Index (CI):**

$$CI = \frac{\lambda_{\max} - n}{n - 1} = \frac{8.084 - 8}{8 - 1} = \frac{0.084}{7} = 0.0120$$

4. **Calculate Consistency Ratio (CR):**

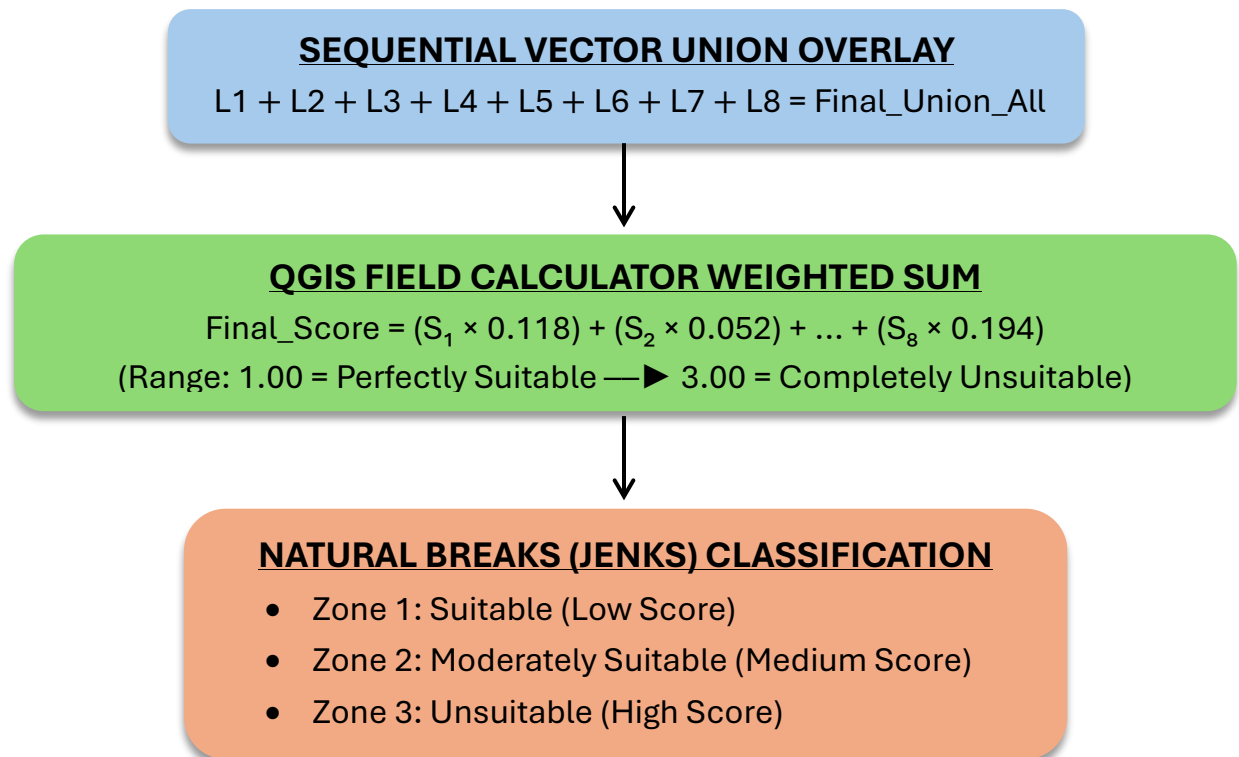
The Random Index (RI) for a matrix of size $n = 8$ is **1.41**.

$$CR = \frac{CI}{RI} = \frac{0.0120}{1.41} = \mathbf{0.0085}$$

Since $CR = \mathbf{0.0085} \ll \mathbf{0.10}$, the pairwise comparisons exhibit excellent consistency, confirming that the derived weights are mathematically valid for spatial modeling.

3.2 Weighted Overlay, Field Calculator Formula, and Final Mapping

Once all eight layers were combined into a single master attribute table via seven sequential vector union operations (Final_Union_All.shp), the continuous suitability score for each unique polygon was computed.



3.2.1 Handling Null Values with coalesce()

During the spatial union process, polygon fragments created outside buffer extents produced NULL values in certain score columns. To prevent NULL propagation from corrupting the final calculation, the coalesce() function was applied inside the equation.

- **Proximity-Beneficial Variables (L1, L2, L3, L4, L5, L6, L7):** Defaulted to 3 (Unsuitable if uncharacterized).
- **Flood Risk (L8):** Defaulted to 1 (Suitable if outside the stream buffer, meaning high, safe ground).

3.2.2 Exact QGIS Field Calculator Formula

Open the **Field Calculator** on your merged layer (Final_Union_All.shp), create a new field named Final_Score (Output Field Type: **Decimal number / real**, Precision: **3**), and paste the following expression:

```
(coalesce("Score_L1", 3) * 0.118) +
(coalesce("Score_L2", 3) * 0.052) +
(coalesce("Score_L3", 3) * 0.063) +
(coalesce("Score_L4", 3) * 0.165) +
(coalesce("Score_L5", 3) * 0.055) +
(coalesce("Score_L6", 3) * 0.048) +
(coalesce("Score_L7", 3) * 0.305) +
(coalesce("Score_L8", 1) * 0.194)
```

Figure 17: *Final Weighted Sum — QGIS Field Calculator Expression*

3.2.3 Final Map Classification and Zonation

The output column Final_Score yields continuous values ranging between **1.000** (Optimal) and **3.000** (Unsuitable).

To translate these numerical scores into cartographic zones for planning maps, the layer is styled in QGIS using **Graduated Symbolology** under **Natural Breaks (Jenks)** classification into three discrete suitability classes:

Final Score Range	Suitability Zone Class	Color Symbol	Planning & Management Description
1.00 – 1.67	Suitable (Class 1)	Green	Low-hazard, gently sloped terrain with excellent water and scenic access. Primary target zones for low-impact ecotourism campsites.
1.67 – 2.33	Moderately Suitable (Class 2)	Yellow	Usable terrain requiring minor site prep, weather monitoring, or seasonal restrictions due to moderate slope or fringe hazard exposure.
2.33 – 3.00	Unsuitable (Class 3)	Red	High landslide risk, flash flood zones, or steep cliffs. Strictly restricted from any campsite development.

Table 23: *Final Suitability Classification Scheme (Equal Interval)*

Camping Sites Suitability Map Knuckles Conservation Forest, Sri Lanka

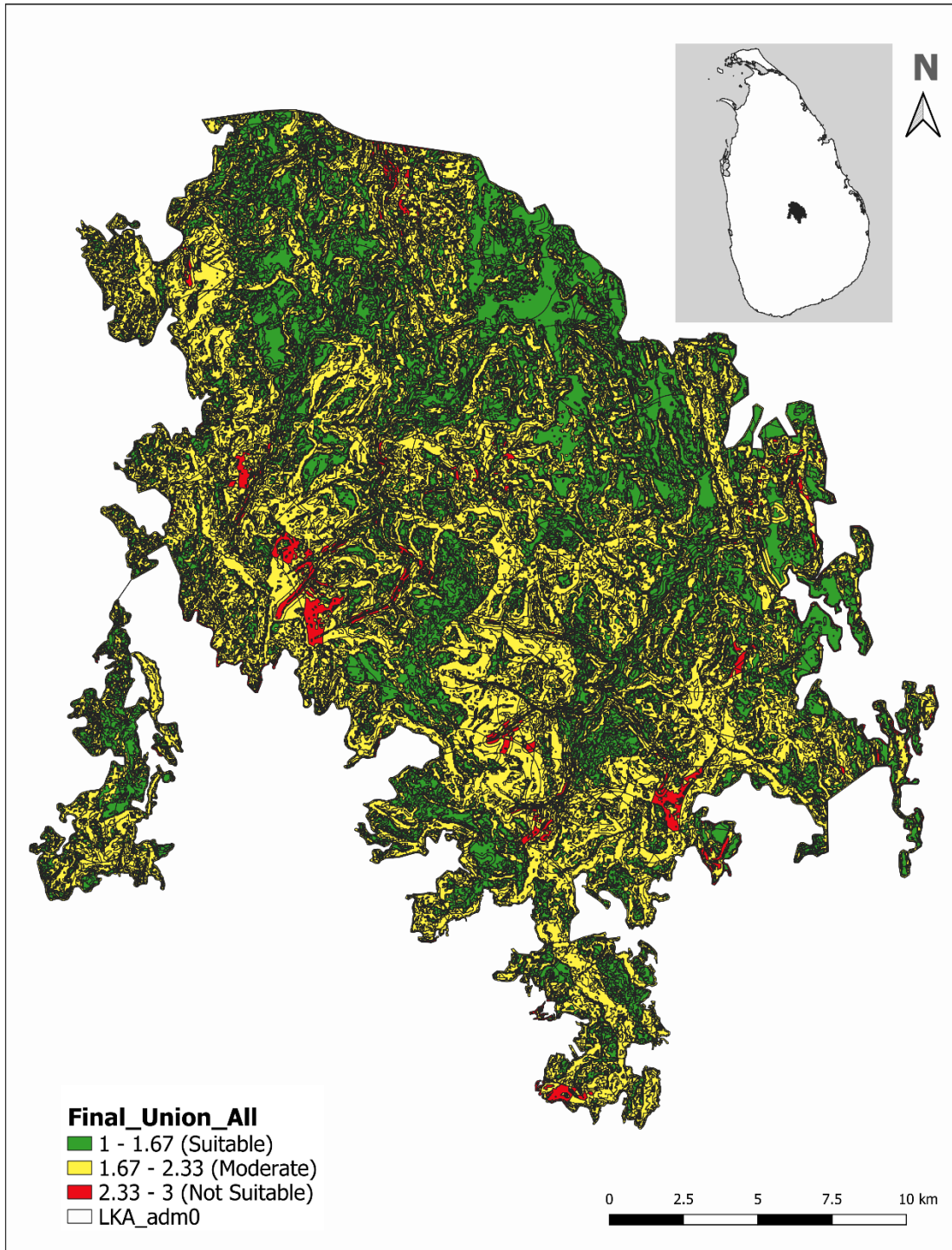


Figure 18: *Final Map - Camping Sites Suitability in Knuckles Conservation Forest, Sri Lanka*

04. Results and Discussion

This section discusses the observed spatial suitability patterns across the Knuckles Conservation Forest study area based on the scoring thresholds and the multi-criteria weighted overlay of all eight spatial variables ($L1 - L8$). Continuous composite scores across `Final_Union_All.shp` range from 1.00 (Optimal Suitability) to 3.00 (Complete Unsuitability). Using Equal Interval classification in QGIS, the output map is categorized into three management classes: **Suitable** (1.00 – 1.67), **Moderate** (1.67 – 2.33), and **Not Suitable** (2.33 – 3.00).

4.1 Natural Environment Suitability

Natural Environment Condition combines Slope ($L1$), Canopy Density ($L2$), and Distance to Waterways ($L3$), carrying a combined AHP weight of **0.234**. The Knuckles range is characterized by steep, highly dissected terrain in its southern and central sections and gentler ground in the north. Consequently, slope-suitable areas ($< 10^\circ$) are concentrated along ridgelines and valley floors rather than the steep escarpments that dominate much of the range.

Canopy Density exhibits broad Not Suitable coverage across the dense montane cloud forest covering most of the interior, with Suitable and Moderate scores confined to patana grassland, rock outcrops, and open/managed land. Suitability for Distance to Waterways is concentrated in narrow bands (100 – 200 m) along the drainage network, balancing water availability against immediate flash flood exposure.

4.2 Landscape Suitability

Landscape Condition is represented solely by Distance to Scenic Spots ($L4$), carrying the second-highest weight (**0.165**) among the eight variables. This reflects that scenic access is the primary visitor motivation for eco-camping in the Knuckles range. Suitable areas cluster within 1.5 km of recorded viewpoints, waterfalls, and peaks. Because OpenStreetMap point-of-interest data is concentrated in specific pockets rather than evenly distributed, Landscape Condition acts as a spatial constraint that restricts rather than expands the Suitable footprint.

4.3 Safety Suitability

Safety Condition variables like comprising Geological Disaster Risk ($L7$) and Flood Risk ($L8$) are carries the largest combined AHP weight (**0.499**) and serves as the primary spatial control on the final map. Given that the Knuckles range receives 2,000 – 3,500 mm of rainfall annually and includes slopes exceeding 30° in its southern reaches, substantial areas fall into the Not Suitable class for Geological Disaster Risk. Flood Risk (scored with reversed distance logic) excludes a further 100 m buffer along the drainage network. Because Safety Condition accounts for nearly half of the composite weight, high hazard risk strictly overrides favorable landscape or infrastructure scores.

4.4 Infrastructure Suitability

Infrastructure Condition combines Telecommunication (L5) and Roads/Transportation (L6), weighted at **0.103** (the lowest of the four groups). Much of the Knuckle's interior is accessible only on foot, and OpenStreetMap network coverage is sparse away from the range's periphery. As a result, Infrastructure Condition acts as a secondary filter, removing remote interior zones from consideration even where natural and safety attributes are favorable.

4.5 Comprehensive Campsite Suitability

The final composite map combines all eight variables into a unified spatial assessment. Table 8 defines the three suitability zones, their assigned map colors, recommended management actions, and structured area statistics placeholders.

Zone Class	Extent Area (km2)	Area Share (%)	Meaning & Physical Profile	Recommended Action
Suitable (1.00 – 1.67)	225.54	44.37%	Gentle terrain, low hazard risk, good scenic access, and close infrastructure proximity.	Recommend for eco-camping development, subject to detailed site review.
Moderate (1.67 – 2.33)	250.16	49.22%	Mixed profile — favorable features offset by moderate slope or limited infrastructure access.	Requires site-specific assessment and minor site preparation prior to development.
Not Suitable (2.33 – 3.00)	32.58	6.41%	High landslide/flood risk, steep scarp terrain (> 30°), dense primary forest, or extreme remoteness.	Strictly restricted from campsite development; preserve for conservation.
Total Area (1.00 – 3.00)	508.28	100.0%	Complete Knuckles Conservation Forest study extent.	

Table 24: Definition, Area Statistics, and Recommended Management Actions for Suitability Zones

Based on the final weighted overlay calculation across the **508.28 km²** study extent, **Moderate (Yellow)** suitability represents the largest proportion of the Knuckles landscape at **250.16 km² (49.22%)**, reflecting extensive transitional terrain across the reserve's plateau margins and intermediate slopes. **Suitable (Green)** zones account for **225.54 km² (44.37%)**, demonstrating substantial opportunity for low-impact eco-tourism development in stable, accessible regions. **Not Suitable (Red)** areas cover **32.58 km² (6.41%)** of the range, effectively isolating high-risk steep scarps, active flash flood zones, and ecologically sensitive core forest from development.

4.6 Sensitivity and Uncertainty

- **Missing Variables:** Forest Fire Risk and Threat of Dangerous Animals were excluded due to local data limitations; their weight was redistributed proportionally across remaining variables.
- **Slope Threshold Sensitivity:** The Suitable/Moderate boundary at 10° (*L1*) is adjustable; relaxing this threshold to 15° would expand the Suitable zone footprint in plateau transition areas.
- **OSM Data Completeness:** Remote interior areas of the forest may have incomplete road or telecommunication features in OpenStreetMap, which could understate infrastructure accessibility (*L5, L6*).
- **DEM Resolution:** The 30 m SRTM pixel size introduces an estimated ± 30 m positional uncertainty along ridge boundaries and steep drop-offs, affecting *L1* and *L7*.
- **Classification Sensitivity:** Class break values (1.67 and 2.33) are tied to the equal-interval range and should always be reported alongside the map legend.

4.7 Stakeholder Implications

Stakeholder Group	Primary Need	Most Relevant Model Output	Suggested Action / Improvement
Department of Wildlife Conservation (DWC)	Identify ecologically sensitive and high-hazard areas to protect.	Not Suitable (Red) zones (dense forest, high landslide/flood risk).	Overlay output with core reserve boundaries to enforce strict non-development zones.
Eco-Tourism Planners	Viable, scenic campsite locations with infrastructure access.	Suitable (Green) zones located near viewpoints and access roads.	Extract top candidate polygon clusters with coordinates and accessibility rankings.
Government Planning Authorities	Spatially accurate outputs compatible with national GIS frameworks.	Projected vector shapefiles (EPSG:5234).	Integrate area statistics (km ² and %) directly into regional master plans.

Table 25: Relevance of the Campsite Suitability Model to Key Stakeholders

4.8 Ethical Considerations

- **Community Impact:** Villages surrounding the Knuckles range rely on forest resources for livelihoods; any site selection must involve local community consultation and benefit-sharing.
- **Open Data Licensing:** OpenStreetMap data used in this study is subject to the Open Database License (ODbL), requiring proper attribution in planning documentation.
- **Conservation Risk:** Publishing suitability maps could inadvertently encourage illegal or unmanaged camping; a "Suitable" tag reflects spatial suitability, not regulatory permission.
- **Hazard Communication:** Moderate zones carry meaningful terrain risks and must be communicated responsibly to planners, rather than treated as unconditionally safe areas.

05. Conclusion

This study developed an open-source, reproducible GIS-based Multi-Criteria Decision Analysis (GIS-MCDA) framework for campsite suitability in the Knuckles Conservation Forest, Sri Lanka. Eight spatial variables spanning Natural Environment, Landscape, Infrastructure, and Safety Condition were standardized to a 1–3 scoring system and integrated via Analytic Hierarchy Process (AHP) weights adapted from Wang et al. (2016).

1. **Safety Dominance:** Safety Condition (*L7* Geological Hazard + *L8* Flood Risk) is the primary spatial determinant, accounting for **49.9%** of the composite decision weight. This confirms that hazard mitigation must precede tourism development in steep, high-rainfall tropical mountain environments.
2. **Methodological Reproducibility:** The workflow demonstrates that complex spatial planning can be executed using entirely free tools and open data (QGIS, SRTM DEM, OpenStreetMap), making the approach adaptable to other protected areas in developing regions.

Limitations & Future Work

The main limitation of this study is the exclusion of Forest Fire Risk and Wildlife Conflict datasets due to localized data gaps. Future research should integrate higher-resolution elevation data (e.g., 10 m Copernicus DEM or LiDAR) to refine slope boundaries, incorporate dynamic rainfall forecasting for real-time hazard alerts, and explore machine learning algorithms (e.g., Random Forest) to capture complex, non-linear spatial interactions.

Ultimately, the resulting suitability map and zonation provide an evidence-based spatial tool for the Department of Wildlife Conservation and eco-tourism planners, balancing sustainable recreational development with ecological protection.

06. References

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